

IE500 Data Mining | University of Mannheim

Team 04 · Adult Income Prediction

1. Work Plan & Team Responsibilities

Each task is assigned to a specific sub-team for balanced workload and peer review. Deliverables reference sections of the Revised Project Outline (submitted April 15, 2026). Status is updated after each sprint. All members must attend at least one Thursday coaching session.

#	Date Range	Task / Role	Members	Deliverable (ref. Revised PO, Apr 15 2026)	Status
1	April 15, 2026	Revised Project Outline Submitted	All Members	- Revised 4-page outline submitted to Prof. Dr. Bizer via Ilias. - Covers: Problem Definition, Dataset Description, Solution Approach (3.1–3.4), Evaluation Method, Interpretation Plan, Work Plan.	✓ Done
2	April 23–24, 2026	Data Acquisition, EDA & Visualizations (Sec. 2 – Dataset Description)	Hazar + Nabin + Xiaowei	- EDA notebook with: class distribution (76.1% / 23.9%), demographic breakdowns, correlation heatmap. - Missing value audit: workclass 3.76%, occupation 3.77%, native-country 1.19%. - Single-feature baseline analysis (capital-gain, hours-per-week).	↻ In Progress
3	April 25–28, 2026	Preprocessing Pipeline & Feature Engineering (Sec. 3.1–3.3)	Hazar + Nabin + Xiaowei	- Mode imputation for missing values; OHE for low-cardinality features. - Target Encoding for native-country; StandardScaler on numeric features. - Engineered columns: capital_net, hours_band (4 tiers), education_level (4 tiers). - Output: train / val / test CSVs saved to GitHub.	⌚ Pending
4	April 29–May 2, 2026	Baseline & Symbolic Models + Evaluation Setup (Sec. 3.4 & Sec. 4)	Namrath + Pravallika + Raywin	- Models: Majority Class Baseline, Logistic Regression (with coefficients), Decision Tree (max_depth=5, visualised). - Evaluation: F1-Score (primary), ROC-AUC (secondary), Accuracy, Precision, Recall, Confusion Matrices. - Stratified 80/20 train-val split; results logged to central Google Doc.	⌚ Pending

#	Date Range	Task / Role	Members	Deliverable (ref. Revised PO, Apr 15 2026)	Status
5	May 3–4, 2026	Hyperparameter Tuning & Cross-Validation (Sec. 4 – Split Strategy)	Nabin + Hazar	5-fold Stratified K-Fold CV on training set. - GridSearchCV / RandomizedSearchCV tuning logs for RF and XGBoost. - Full metric table (all models) vs. UCI benchmark; official UCI test-set holdout locked.	 Pending
6	May 5–9, 2026	Advanced Models, Ensemble & Interpretation (Sec. 3.4 Advanced & Sec. 5)	Namrath + Pravallika + Raywin	- Models: SVM (RBF), Random Forest (n=200), XGBoost (n=300); Soft Voting Ensemble (LR + RF + XGB). - Interpretation: LR coefficient plot, Decision Tree rules, RF/XGBoost feature importance (Gini/Gain). - Fairness analysis across sex, race, and native-country.	 Pending
7	May 10–16, 2026	Final Project Report (12-page DWS LaTeX template)	All Members	- Max. 12 pages (10 content + title/TOC + references). No appendix. - Sections: Application area & goals, Dataset profile, Preprocessing, Data Mining, Evaluation (error analysis + benchmark comparison), Results. - Declaration of AI tools used. Python code + data subset uploaded to Ilias by May 17, 23:59.	 Pending
8	May 17–20, 2026	Final Presentation Slides (10-min talk + 5-min Q&A)	All Members	- PDF slides: problem definition, dataset, preprocessing decisions, model comparison table (all 6 models). - Feature importances, fairness findings, conclusion & limitations. - Upload to Ilias by May 21, 23:59. All members attend assigned 90-min session.	 Pending

Note: Each extra report page or day of late submission downgrades the mark by 0.3. AI tool usage (e.g. Claude) must be declared in the final report using the Declaration Table format.

2. Infrastructure

The team uses a shared, version-controlled infrastructure to ensure reproducibility, transparent result tracking, and smooth collaboration across all sprints. All experiment results are logged with the exact GitHub commit hash that produced them.

#	Tool / Platform	Purpose & Usage	Responsible
A. Code & Version Control			
A1	GitHub	Version control for all notebooks, preprocessing scripts, modelling code, and data subsets. One branch per pipeline stage; merge to main after peer review.	All members — Hazar (repo admin)
A2	VS Code / Jupyter	Primary development environment. One Jupyter notebook per stage: eda.ipynb, preprocessing.ipynb, modelling.ipynb, evaluation.ipynb.	All members
B. Python Libraries			
B1	Core stack	pandas, numpy - data loading, wrangling, and feature engineering.	All members
B2	ML & Evaluation	scikit-learn (LR, DT, RF, SVM, GridSearchCV, metrics), xgboost (XGBClassifier), category_encoders (TargetEncoder).	All members
B3	Visualisation	matplotlib, seaborn - EDA plots, confusion matrices, ROC curves, feature importance charts.	All members
C. Communication & Coordination			
C1	MS Teams / Zoom/Whatsapp	Weekly sprint sync (Thursdays after coaching session). Quick async questions and file sharing within the team.	All members — Nabin (meeting facilitator)
C2	Ilias Message Board	Official submissions (project outline, report, slides). Coaching session sign-up. Announcements from Prof. Dr. Bizer.	All members
D. Documentation & Results Tracking			
D1	Google Doc (shared)	Central experiment log: model name, hyperparameters, F1/AUC on val & test, GitHub commit hash that produced the result. Updated after every completed sprint.	All members
D2	LaTeX (Overleaf)	Final 12-page project report using DWS master thesis template. Sections assigned per sprint owner; merged for final submission.	All members
D3	Google Slides / PDF	Final 10-minute presentation slides exported to PDF and uploaded to Ilias by May 21, 23:59.	All members

Workflow rule: No result is recorded in the Google Doc without a corresponding committed notebook on GitHub. Branches follow the naming convention: stage/eda, stage/preprocessing, stage/modelling, stage/evaluation.